Week 1 — Introduction to MySQL & database design (ER)

Project README · Week 2 →

Lab: Handout lab list **#1 — Introduction to MySQL and ER Diagram** · lab manual **Experiment 0/1 — Introduction to SQL & MySQL and Database Design Theory lectures this lines up with:** 1–9 (introduction to databases, ER modelling, ER → relational design)

Design before you build. You create the empty `sentinel` database and, more importantly, write down the problem, the requirements, the ER diagram and the rules that turn that diagram into tables. No table exists yet.

What was added

File	What it is
`sql/01_create_database.sql`	`DROP/CREATE DATABASE sentinel`, utf8mb4. Every later week starts from this.
`docs/problem-statement.md`	Handout steps 1-4: title, problem, objectives, functional + non-functional requirements, the six fraud signals.
`docs/er-diagram.md` · `er-diagram.dbml`	The ER diagram (Mermaid, renders on GitHub) and DBML for dbdiagram.io. **Generated from the real schema**, never hand-edited.
`docs/er-to-relational.md`	Which ER-to-tables rule produced which table (strong entity, 1:N, M:N, 1:1, participation, derived attribute).
`demo/01_mysql_basics.sql`	`SHOW DATABASES`, `USE`, `SHOW TABLES`, `VERSION()`: the commands in the lab manual's intro page.

Files in `sql/` are **loaded by the build** (`scripts/build.sh`). Files in `queries/`, `demo/` and `lab-practice/` are run by hand and are checked by `scripts/check.sh`.

What you should see

- `SHOW DATABASES` lists `sentinel`; `SHOW TABLES` is empty (tables arrive in Week 2).
- Character set `utf8mb4`, collation `utf8mb4_unicode_ci`.

Concepts you must be able to explain (viva)

- **DBMS vs file system.** A DBMS adds integrity rules, concurrent access control, security and data independence that plain files lack.
- **The five parts of SQL.** DDL (structure) · DML (change data) · DRL (SELECT) · TCL (COMMIT/ROLLBACK) · DCL (GRANT/REVOKE). The lab manual lists exactly these five.
- **ER vocabulary.** Entity, entity set, attribute (simple/composite/multivalued/derived), relationship, cardinality (1:1, 1:N, M:N), participation (total/partial), weak entity.
- **ER → relational.** Read `docs/er-to-relational.md` twice. Each row is a Part B/C exam answer.
- **Why surrogate keys.** `email`, `fingerprint_hash` are candidate keys; a small `INT` id is the primary key because it is compact in every foreign key and index.

More questions with full answers: `week-11-final-polish/docs/viva_questions.md`. Syntax for the closed-book lab: `docs/LAB_EXAM_CHEATSHEET.md`.

Handout step covered

Handout steps 1-5 (title, problem, objectives, requirements, ER model).

Results (captured from the running database)

Evidence: Database and character set

Script: `week-01-intro-er-design/demo/01_mysql_basics.sql`

```
+--------------+
| mysql_version |
+--------------+
| 8.4.11        |
+--------------+
+--------------------+
| Database           |
+--------------------+
| information_schema |
| mysql              |
| performance_schema |
| sentinel           |
| sys                |
+--------------------+
+------------------+
| current_database |
+------------------+
| sentinel         |
+------------------+
+------------------------+
| Tables_in_sentinel     |
+------------------------+
| account_devices        |
| accounts               |
| alerts                 |
| audit_log              |
| behaviour_baselines    |
| devices                |
| locations              |
| login_history          |
... (22 more lines)
```